



SUBJECT: Computer Systems 4A (RKEEEA4)

TITLE: Practical Assessment 1 and 2

DATE: 17 March 2026

Introduction to smart sensors

The purpose of these practicals are to introduce students to design and build basic smart sensors which allow for users to remote monitor sensor data, as well as allow for remote control of actuators.

You are allowed to work in groups, no larger than 3, and 1 practical report shall be submitted for the group.

The practical report shall contain the usual discussion of the practical, as well as answer the questions which will be provided later on. 1 combined practical report for both practical 1 and 2 shall need to be submitted, 1 week after the practical 2 demonstrations take place.

Practical 1: Smart sensors

Students are required to build a smart sensor that allows for the reading of sensor data on a webpage.

Equipment given:

- Wi-Fi equipped microcontroller, either an ESP32 or Arduino Nano ESP32
- Analogue sensor, either DHT11, MQ3 or similar.

What will need to be demonstrated:

The practical will be carried out in multiple stages, to demonstrate understanding on how to program microcontrollers.

- 1) Demonstrate the ability to blink the onboard LED on the microcontroller.
- 2) Demonstrate the ability to display the sensor data locally
- 3) Demonstrate the ability to display the sensor data on a webpage.

At each demonstration you will be asked questions on the code used and be could be asked to change the code to show understanding of how it works, for example you could be asked to change the pattern of the blinking on board LED.

Once all 3 demonstrations are completed each group will be asked 5 questions about their smart sensor, or about networking in general.

Mark structure:

The final mark for practical one shall be calculated as follows:

50% demonstration mark + 50% Report mark.

The demonstration mark shall be calculated as follows:

40% build and coding demonstrations + 60% questions answered.

Practical 2: Smart sensors continued.

Students are required to control an actuator using a webpage.

Equipment given:

- Wi-Fi equipped microcontroller, either an ESP32 or Arduino Nano ESP32 (please use a different controller to the one your group has used in practical 1.)
- Actuator, either a fan, servo or similar.

What will need to be demonstrated:

The practical will be carried out in multiple stages, to demonstrate understanding on how to program microcontrollers.

- 1) Demonstrate the ability to blink the onboard LED on the microcontroller.
- 2) Demonstrate the ability to control the actuator locally
- 3) Demonstrate the ability to control the actuator through a webpage.

At each demonstration you will be asked questions on the code used and be could be asked to change the code to show understanding of how it works, for example you could be asked to change the pattern of the blinking on board LED.

Once all 3 demonstrations are completed there each group will be asked 5 questions about their smart sensor, or about networking in general.

Mark structure:

The final mark for practical one shall be calculated as follows:

50% demonstration mark + 50% Report mark.

The demonstration mark shall be calculated as follows:

40% build and coding demonstrations + 60% questions answered.

The following are the questions that need to be answered in the practical report.

- 1) Explain all the hardware devices in the network you are utilising and what their function(s) are. This must include end devices and networking devices. (It is understood that the exact network architecture is not known, but you can provide generic assumptions)
- 2) Draw a picture of the network architecture including the network hardware discussed in (1), starting at the Arduino, moving to the cloud platform and back to be viewed on your computer. Show the path of the information from the Arduino to your computer screen.

- 3) For every hop drawn in the figure in (2), add the TCP/IP layers which were active during every hop. Show which TCP/IP layers and indicate how the packets are encapsulated and decapsulated during the movement from the Arduino to your computer screen.
- 4) Why do you not see the results from your sensor instantly on your PC? Explain the delay?
- 5) How does the microcontroller used get its IP address? Explain.
- 6) Discuss the software used in this practical. Discuss the function of every piece of software.